

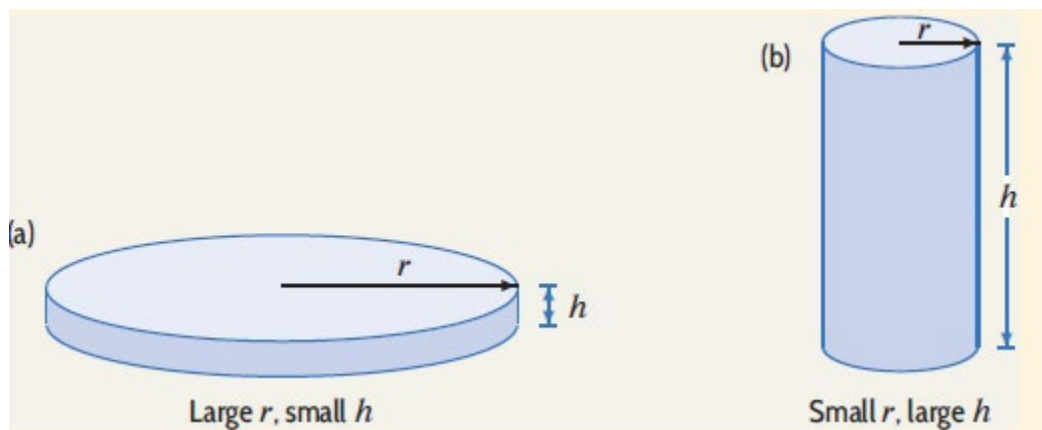
Sec 4.3 – Optimization and Modeling

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

In Sec 4.2 we found **global extrema** of a function we were given. When we translate a (“word”) problem into a function with a known formula, this is called **mathematical modeling**. Once we have the model, or function, we can use the techniques from Sec 4.2 to find the global extrema.

What are the dimensions of an aluminum can that holds up to 40 in^3 of juice and uses the **least material**?

We will assume the can is cylindrical and capped at both ends.



If we try to save material by making the top and bottom of the can small, then the can has to be tall to accommodate all 40 in^3 of juice.

We'll start by using modeling to set up the equation to minimize the material.

M = the material in the can = the material in each end + the material in the side.

Material in the ends = $2 \cdot \text{Area of a circle with radius } r, \rightarrow 2 \cdot \pi \cdot r^2$

Material in the side = Area of the curved side of the cylinder with height h and radius $r, \rightarrow 2 \cdot \pi \cdot r \cdot h$
(think of cutting the side, unwrapping it, and laying it flat).

This gives us our equation for the Material that we want to minimize: $M = 2\pi r^2 + 2\pi r h$

But, we have two variables, r and h . These variables are related, as we saw above, when one is small, the other is large.

In particular, they are related by a constraint, that the can has to hold 40 in^3 of juice.

$V = \pi r^2 h \rightarrow 40 = \pi r^2 h$. We solve for h and substitute into our equation to minimize.

$h = \frac{40}{\pi r^2} \rightarrow M = 2\pi r^2 + 2\pi r \frac{40}{\pi r^2} \rightarrow M = 2\pi r^2 + \frac{80}{r}$ and $r > 0$, it doesn't make physical sense otherwise.

To minimize M , we take the derivative, $\frac{dM}{dr} = 4\pi r - \frac{80}{r^2}$ and then set it equal to 0 to find the critical points (c.p.),

$0 = 4\pi r - \frac{80}{r^2} \rightarrow 4\pi r = \frac{80}{r^2} \rightarrow r^3 = \frac{80}{4\pi} \rightarrow r = 1.85$ inches. We substitute back to find h ,

$h = \frac{40}{\pi (1.85)^2} = 3.7$ inches.

We confirm that we have the global minimum, we consider the formula for $\frac{dM}{dr}$.

For small r , $r < 1.85$, the $-\frac{80}{r^2}$ term dominates, and $\frac{dM}{dr} < 0$, and for large r , $r > 1.85$, the $4\pi r$ term dominates and $\frac{dM}{dr} > 0$.

So M is decreasing for $r < 1.85$ and M is increasing for $r > 1.85$, so the global min is at $r=1.85$.

Now, we will work some problems together. The solution to these problems, and other problems, are linked in Moodle.

Practical Tips for Modeling Optimization Problems

1. Make sure that you know what quantity or function is to be optimized.
2. If possible, make several sketches showing how the elements that vary are related. Label your sketches clearly by assigning variables to quantities which change.
3. Try to obtain a formula for the function to be optimized in terms of the variables that you identified in the previous step. If necessary, eliminate from this formula all but one variable. Identify the domain over which this variable varies.
4. Find the critical points and evaluate the function at these points and the endpoints (if relevant) to find the global maxima and/or minima.

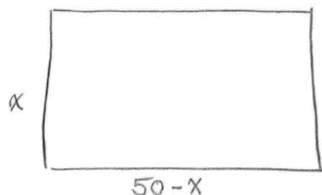
Partner up

Example 1 Sam wants to build a fenced rectangular garden and has 100 feet of fence. Sam wants the garden to have the maximum possible area; find the dimensions of such a garden.

Strategies:

- Draw a picture of the situation, sometimes more than one picture to see how it changes.
- Give a variable to any quantities that are changing.
- Write the quantity you want to optimize in terms of the variables.
- Write any constraint equations (relationships between the variables) and use them to simplify the quantity you want to optimize to a function of one variable.
- Find the domain of the variable.
- Optimize: find critical points and use one of our methods to determine where the global maximum or minimum is.

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1. Draw a picture of the garden.



2. Let x be the length of one side of the garden. Label your picture with x and the

3. What is the largest value x can be? What is the smallest?

$$0 \leq x \leq 50$$

4. Write a function for the area of the garden.

$$A(x) = x(50-x)$$

5. Find the maximum value of the area function on the interval you gave in #3.

$$\begin{aligned} A(x) &= 50x - x^2 \\ \frac{dA}{dx} &= 50 - 2x = 0 \\ 50 &= 2x \\ 25 &= x \end{aligned}$$

The maximum area of the garden is 625 ft^2 , with dimensions $25 \text{ ft} \times 25 \text{ ft}$

Check values at endpoints & critical pts

$$A(0) = 0$$

$$A(25) = 625 \leftarrow \text{global max}$$

$$A(50) = 0$$

(Rather than check endpoints you could

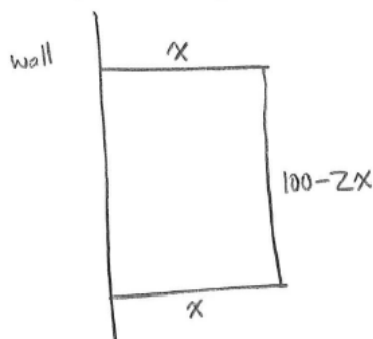
- $A(x)$ is a downward facing parabola
- OR • $A''(x) = -2$, so the whole graph is concave down

Example 2 Suppose Sam decides to build the garden with one side against an existing wall. He still has 100 feet of fence. Find the dimensions of the garden with maximum area in this situation.

One way to solve:

1. Draw a picture of the garden with the wall acting as one side.
2. Let x be the length of one side of the garden. Label your picture with x and the other side length(s).
3. What is the largest value x can be? What is the smallest?
4. Write a function for the area of the garden.
5. Find the maximum value of the area function on the interval you gave in #3.

- =====
1. Draw a picture of the garden with the wall acting as one side.



2. Let x be the length of one side of the garden. Label your picture with x and the other side length(s).
3. What is the largest value x can be? What is the smallest?

$$0 \leq x \leq 50$$

4. Write a function for the area of the garden.

$$A(x) = x(100 - 2x)$$

5. Find the maximum value of the area function on the interval you gave in #3.

$$A(x) = 100x - 2x^2$$

$$A'(x) = 100 - 4x$$

$$100 - 4x = 0$$

$$x = 25$$

Check endpoints/critical pts:

$$A(0) = 0$$

$$A(25) = 1250 \leftarrow \text{global maximum}$$

$$A(50) = 0$$

The garden with maximum area is 25 ft x 50 ft,
with area of 1250 ft²

Example 4 Find two nonnegative numbers whose sum is 9 and so that the product of one number and the square of the other number is a maximum.

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Example 4. Find two nonnegative numbers whose sum is 9 and so that the product of one number and the square of the other number is a maximum.

x, y nonnegative numbers

$$0 \leq x, y \leq 9$$

$$\text{Constraint: } x + y = 9 \Rightarrow x = 9 - y$$

$$\text{Optimize } x \cdot y^2 = (9 - y)y^2 = f(y)$$

* can also solve for $y = 9 - x$ in constraint, but this gives a function with more work to find the derivative: $x(9 - x)^2$

$$f(y) = 9y^2 - y^3$$

$$f'(y) = 18y - 3y^2$$

$$18y - 3y^2 = 0$$

$$3y(6 - y) = 0 \quad y = 0, 6$$

Check endpoints/critical pts:

$$f(0) = 0$$

$$f(6) = 9(6)^2 - 6^3 = 108$$

$$f(9) = 0$$

So the maximum product is for the numbers $3 \frac{1}{2}$ and 6, with product $3 \cdot 6^2 = 108$.